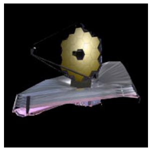

7433 - Mass inflow structures in star formation from filaments to streamers: the unique field of Orion Molecular Cloud 3

Cycle: 4, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
Dr. Mika Juvela (PI) (ESA Member)	**University of Helsinki**
Dr. Dana Alina (CoI)	Nazarbayev University
Dr. Karine Demyk (CoI) (ESA Member)	Institut de Recherche en Astrophysique et Planetologie
Tie Liu (CoI)	Chinese Academy of Sciences
Emma Mannfors (CoI) (ESA Member)	University of Helsinki
Dr. Deborah Paradis (CoI) (ESA Member)	Institut de Planetologie et d'Astrophysique de Grenoble
Dr. Veli-Matti Pelkonen (CoI) (ESA Member)	Universitat de Barcelona
Dr. Isabelle Ristorcelli (CoI) (ESA Member)	Institut de Recherche en Astrophysique et Planetologie
Dr. L. Viktor Toth (CoI) (ESA Member)	Eotvos Lorand University
Mr. Jakobus Marthinus Vorster (CoI) (ESA Member)	University of Helsinki
Dr. Nathalie Ysard (CoI) (ESA Member)	Institut d'Astrophysique Spatiale

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	2	OMC3-M	MIRI Imaging	(2) OMC3-M
	3	OMC3-M-BG	MIRI Imaging	(4) MIRI_BG
	1	OMC3-N	NIRCam Imaging	(1) OMC3-N
	11	Residual Persistence Check	NIRCam Dark	NONE

ABSTRACT

We propose to image the Orion Molecular Cloud 3 (OMC-3), to trace mass transfer along the star-formation (SF) process. Mid-infrared (MIR) extinction will be used to measure column density N(H) at an unprecedented resolution, independent of the large temperature and gas-phase abundance uncertainties that affect ALMA studies. Further aided by NIR-MIR scattering, we can measure in exquisite detail structures from large-scale massive filaments (~0.5 pc field) down to the smallest scales (0.3", ~120 au), over a wide range of column densities, N(H) ~ 1e21 - 1e23 cm-2. Together with the existing line and dust polarization data, this will help to unravel the formation mechanisms of filaments and confirm the the role of fibers in the growth of protostellar cores. Most importantly, JWST data will give an unprecedented view into the mass inflow at the smallest scales, such as the streamers that feed prestellar cores and potentially even young stellar objects (YSOs). The results will have profound impact on our understanding of SF, the evolution of protostellar disks, and the planet formation. Thanks to the wavelength coverage, it is also possible to separate the contributions of light scattering (<5 μm; very sensitivity to grain growth) and thermal dust emission (>10μm; measure of small-grain populations). Together with spatially resolved extinction curves, the data will also lead to full characterization of dust evolution in the field. OMC-3 is a unique target, being located in the closest high-mass SF cloud (d~390 pc), with known strong MIR extinction. The science case is similarly unique to JWST, unable to be addressed with any other past or present instrument.

OBSERVING DESCRIPTION

The measurements consist of NIRCam and MIRI photometric observations of a field in the OMC-3 cloud in Orion. The objectives include the high-resolution imaging of MIR extinction, NIR-MIR scattering, and MIR emission.

The MIRI observations consist of a single pointing that is repeated for four filters: F560W, F770W, F1000W, and F1800W. Multiple filters are needed to ensure a high dynamical range for the mid-infrared (MIR) absorption mapping and for the detection of thermal emission from small dust grains. The extinction observations should be able to probe column densities at a relative accuracy of ~10% in the range N(H)=1e21 - 3×10^23 cm^-2 . The integration times are based on (1) the depth of the mid-infrared absorption that is at low resolution known from Spitzer and WISE observations, and (2) the results from radiative transfer models of the OMC-3 region (matching existing FIR data from Herschel, Artemis and MIR data from Spitzer and WISE) . The requested signal-to-noise ratios exceed 100 per band, when measured relative to the maximum depth of the MIR absorption dip. The shorter wavelengths are more sensitive to optical depth variations at low column densities (~1e21-1e22 cm-2), while the longest wavelengths (10-21 μm) are critical for reaching the highest column densities close to ~1e23 cm-3. The use of multiple filters also provides constraints on the dust extinction curve. The contributions of thermal dust emission becomes significant outside the densest regions. With the proposed sensitivity, we expect to reach SNR~25 for the emission down to very low column densities, for N(H) a few times 1e21 cm^-2.

The NIRCam observations also consist of a single pointing, using two pairs of filters: F150W+F356W and F200W+F480M. Dithering is used to ensure a continuous coverage over an area of 2×5 arcmin. The NIRCam integration times are based on the expected level of scattered light (>~1 MJy/sr) and the existing Spitzer and WISE measurements on the MIR absorption, which will dominate over scattering in the regions of highest column density (the filaments and cores). Figure 2d noise values are given per beam. The NIRCam data will provide an order of magnitude improvement over the Spitzer resolution, with the high sensitivity that is required for the spatial mapping of the scattered light intensity (e.g. the detection of streamers and striations), the estimation of the extinction curve, and the study of the structure of the filaments and cores (outer parts, Av<100 mag) at scales down to angular resolution of ~0.3 arcsec (~120 au at the distance of Orion).

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	OMC3-N	RA: 05 35 24.4000 (83.8516667d) Dec: -05 02 20.00 (-5.03889d) Equinox: J2000	Epoch of Position: 2000	
Comments: Centre coordinates for the NIRCam observation. Depending on the time of the observations and the detector position angle, the coordinates should be adjusted to make sure that the observations cover a maximal area of the filaments SE of the core, and that (as far as possible) also the core G208 is covered while the brightest stars NE of the coverage shown in Fig. 1 are avoided. *Category=ISM* *Description=[Dense interstellar clouds, Filamentary nebulae, Interstellar absorption, Interstellar dust, Protostars]* *Extended=YES*				
(2)	OMC3-M	RA: 05 35 23.8000 (83.8491667d) Dec: -05 01 9.00 (-5.01917d) Equinox: J2000	Epoch of Position: 2000	
Comments: Centre coordinates for the MIRI observation. Depending on the time of the observations and the detector position angle, the coordinates should be adjusted to ensure best coverage of the main filament and, if possible, also of the core G208. *Category=ISM* *Description=[Dense interstellar clouds, Filamentary nebulae, Interstellar absorption, Interstellar dust, Protostars]* *Extended=YES*				
(4)	MIRI_BG	RA: 05 34 38.3000 (83.6595833d) Dec: -04 19 25.10 (-4.32364d) Equinox: J2000	Epoch of Position: 2000	
Comments: *Category=ISM* *Description=[Diffuse interstellar clouds]* *Extended=YES*				

Observation

Proposal 7433, Observation 2: OMC3-M Mon Jan 12 19:00:10 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Comments: The MIRI observing times are based mainly on the expected absorption signal that is already known at lower resolution from the Spitzer/IRAC and WISE observations. The signal-to-noise ratios in the ETC are for the ratio between the depth of the absorption dip and the noise. The foreseen sensitivity for the combination of all selected filters is shown in the proposal, based on direct calculation and based on radiative transfer simulations.

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	OMC3-M	RA: 05 35 23.8000 (83.8491667d) Dec: -05 01 9.00 (-5.01917d) Equinox: J2000	Epoch of Position: 2000	

Comments: Centre coordinates for the MIRI observation. Depending on the time of the observations and the detector position angle, the coordinates should be adjusted to ensure best coverage of the main filament and, if possible, also of the core G208.
Category=ISM
Description=[Dense interstellar clouds, Filamentary nebulae, Interstellar absorption, Interstellar dust, Protostars]
Extended=YES

Template

Subarray
FULL

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	4-Point-Sets				1	1	EXTENDED SOURCE	POSITIVE	DEFAULT
2	CYCLING	1	8						LARGE

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F560W	FASTR1	23	2	1	Dither 2	8	16	1043.415	222533
2	F770W	FASTR1	23	2	1	Dither 2	8	16	1043.415	222533
3	F1000W	FASTR1	22	3	1	Dither 2	8	24	1509.622	222533
4	F1800W	FASTR1	22	3	1	Dither 2	8	24	1509.622	222533

Special Requirements

Group Observations 2, 3, Non-interruptible

Observation

Proposal 7433, Observation 3: OMC3-M-BG

Mon Jan 12 19:00:10 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Comments: The MIRI observing times are based mainly on the expected absorption signal that is already known at lower resolution from the Spitzer/IRAC and WISE observations. The signal-to-noise ratios in the ETC are for the ratio between the depth of the absorption dip and the noise. The foreseen sensitivity for the combination of all selected filters is shown in the proposal, based on direct calculation and based on radiative transfer simulations.

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	MIRI_BG	RA: 05 34 38.3000 (83.6595833d) Dec: -04 19 25.10 (-4.32364d) Equinox: J2000	Epoch of Position: 2000	

Comments:
Category=ISM
Description=[Diffuse interstellar clouds]
Extended=YES

Template

Subarray
FULL

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	4-Point-Sets				1	1	EXTENDED SOURCE	POSITIVE	DEFAULT
2	CYCLING	1	4						LARGE

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F560W	FASTR1	11	1	1	Dither 2	4	4	122.102	222533
2	F770W	FASTR1	11	1	1	Dither 2	4	4	122.102	222533
3	F1000W	FASTR1	8	2	1	Dither 2	4	8	188.703	222533
4	F1800W	FASTR1	8	2	1	Dither 2	4	8	188.703	222533

Special Requirements

Group Observations 2, 3, Non-interruptible

Observation

Proposal 7433, Observation 1: OMC3-N Mon Jan 12 19:00:10 GMT 2026

Diagnostic Status: Error

Observing Template: NIRCam Imaging

Comments: The signal-to-noise ratios in the ETC are based on the expected level of scattered light, which are partly based on previous observations of other targets such as CrA (in a lower radiation field), partly on the Spitzer/IRAC data on the OMC-3 field.

Diagnostics

(OMC3-N (Obs 1)) Error (Form): Permission has not been granted for this program to use value true for field Exclusive Use Of Instrument.

(OMC3-N (Obs 1)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 1:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 1:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.

(Visit 1:2) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.

(Visit 1:3) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.

(Visit 1:4) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	OMC3-N	RA: 05 35 24.4000 (83.8516667d) Dec: -05 02 20.00 (-5.03889d) Equinox: J2000	Epoch of Position: 2000	

Comments: Centre coordinates for the NIRCam observation. Depending on the time of the observations and the detector position angle, the coordinates should be adjusted to make sure that the observations cover a maximal area of the filaments SE of the core, and that (as far as possible) also the core G208 is covered while the brightest stars NE of the coverage shown in Fig. 1 are avoided.
Category=ISM
Description=[Dense interstellar clouds, Filamentary nebulae, Interstellar absorption, Interstellar dust, Protostars]
Extended=YES

Template

Module	Subarray	Target Placement
ALL	FULL	Module gap (large extended source)

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	FULL	6	STANDARD		1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F200W	F480M	BRIGHT1	10	1	6	6	1223.992	222533
2	F150W	F356W	BRIGHT1	10	1	6	6	1223.992	222553

Special Requirements	
Special Requirements	Sequence Visits , Non-interruptible, Exclusive Use Of Instrument Aperture PA Range 84.92542306 to 174.92542306 Degrees (V3 85.0 to 175.0) Aperture PA Range 264.92542306 to 354.92542306 Degrees (V3 265.0 to 355.0) Visits Same PA Persistence Risk 11 After 1 by 4.63 Hours to 2 Days, Exclusive Use Of Instrument

Observation

Proposal 7433, Observation 11: Residual Persistence Check

Mon Jan 12 19:00:10 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRCam Dark

Diagnostics

(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Template

Module	Subarray	Science Template	Occulting Mask	No. of Output Channels
ALL	FULL	Imaging		4

Spectral Elements

#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID
1	MEDIUM8	10	1	1	1	1052.203	

Special Requirements

11 After 1 by 4.63 Hours to 2 Days, Exclusive Use Of Instrument